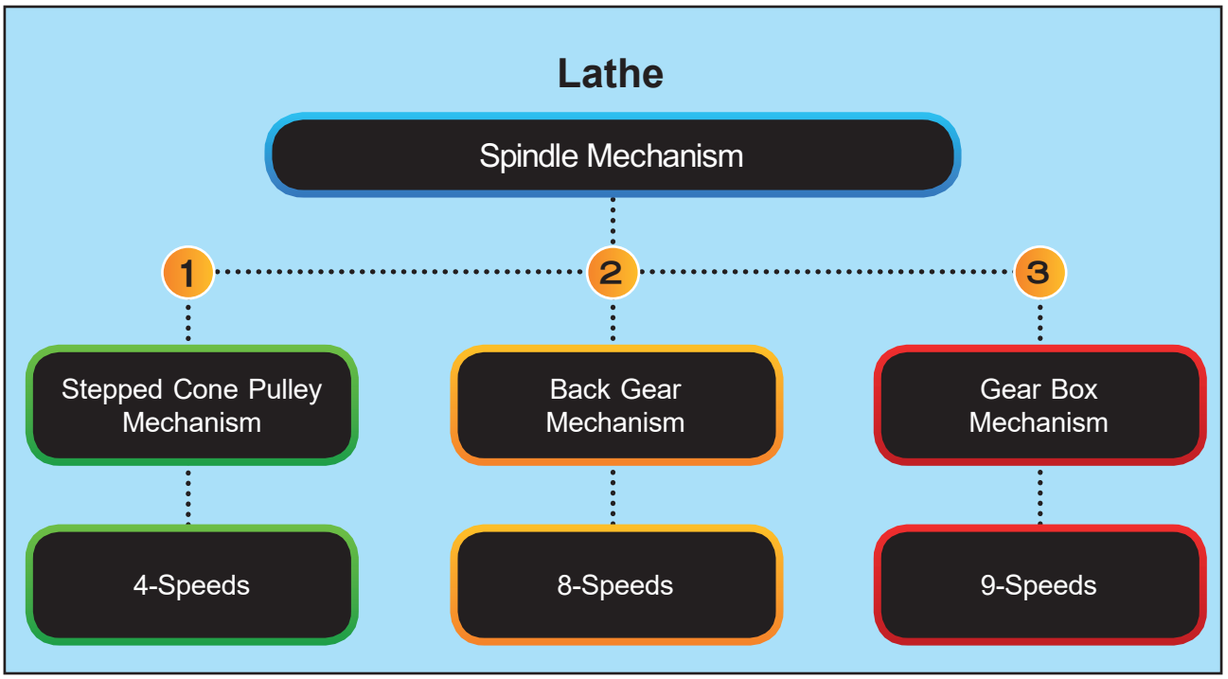
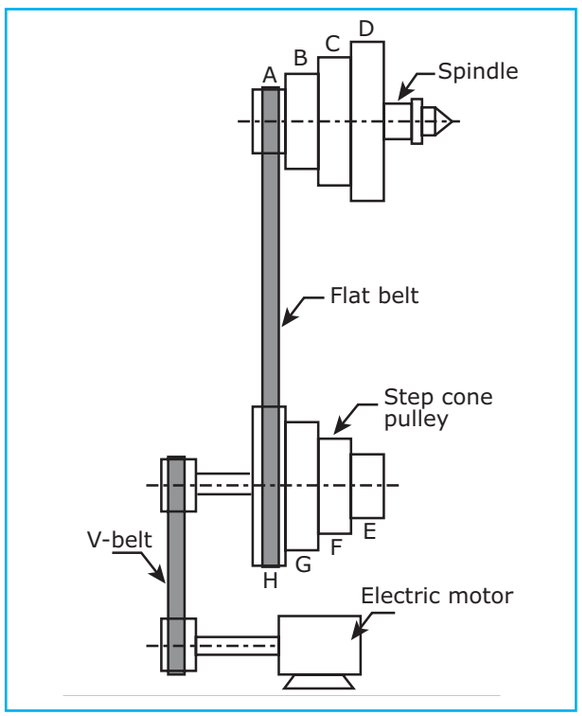


The belt is arranged on any one of the four steps to obtain different spindle speeds. If the belt is placed on the smaller step of the driven pulley, the spindle speed will be increased. For example, when the belt is arranged between A & H the spindle speed will be maximum. When it is between D & E, the speed will be minimum. Stepped cone pulley drive is illustrated in figure.



Stepped Cone Pulley Mechanism

1.10 Back gear mechanism

Back gear mechanism is housed within the headstock of the lathe. A step cone pulley having steps ABCD and a small pinion 'P' are mounted on the spindle and rotates freely on it. The gear 'S' is keyed to the head stock spindle. So the spindle will rotate only when the gear 's' rotates.

The step cone pulley ABCD and the gear 'S' can be kept separately or made as one unit with the help of a pin 'T' when the pin is disengaged, the cone pulley along with the gear P will rotates freely on the spindle and the spindle will not rotate. There is another shaft parallel to the spindle axis having back gear Q and R mounted on it. These back gears can be made to mesh with gears 'P' and 'S' or kept disengaged from them. The spindle can get drive either from the cone pulley or through back gears.

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